

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering ¹
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1 Objective

To implement and analyze K-Means, DBSCAN, and Hierarchical (Agglomerative) clustering on the Human Activity Recognition (HAR) sensor dataset, select hyperparameters via the elbow/silhouette method and a density-based parameter search, and compare the three algorithms using both internal (unsupervised) and external (ground-truth-based) validation metrics.

2 Problem Statement

Given the UCI HAR dataset, where each sample is a 561-dimensional feature vector extracted from smartphone accelerometer/gyroscope signals for one of six activities, the objective is to group the data into clusters using three unsupervised algorithms – without using the activity labels during training – and then evaluate how well the discovered cluster structure aligns with the true activities. Input: 561 numeric time/frequency-domain features per sample. Output: a cluster label per sample from each of the three algorithms, together with internal metrics (Silhouette, Davies–Bouldin, Calinski–Harabasz) and external metrics (Adjusted Rand Index, Normalized Mutual Information) computed against the true activity labels.

3 Dataset Description

Dataset Name	UCI Human Activity Recognition Using Smartphones (HAR)
Dataset Source	UCI Machine Learning Repository
Number of Samples	10,299 (train + test partitions combined)
Number of Features	561 time- and frequency-domain signals derived from 3-axis accelerometer and gyroscope readings (50 Hz, 2.56s windows)
Number of Classes	6 (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) – used only for external evaluation, not for clustering itself
Missing Values	0

Train-Test Split	Not applicable – clustering is unsupervised and was performed on the full combined dataset (original train/test partitions concatenated)
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4 Exploratory Data Analysis

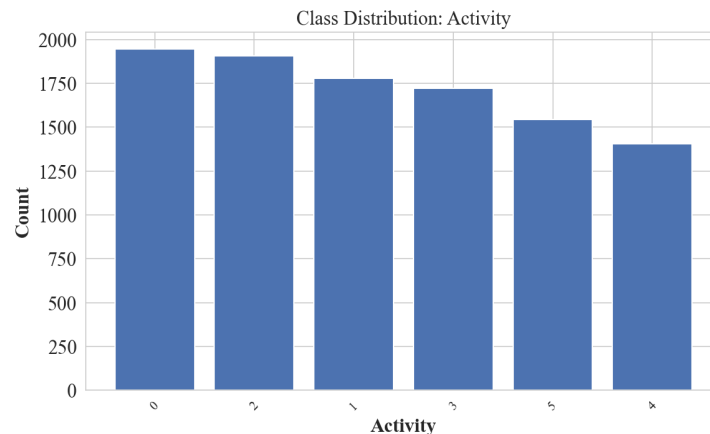


Figure 1: Distribution of the six true activity labels

Inference: The six activities are fairly evenly represented in the combined dataset, with no severe class imbalance. Since clustering does not use these labels during training, this balance mainly affects the external validation step later, ensuring that metrics like ARI and NMI are not dominated by one over-represented activity.

5 Data Preprocessing

- **Missing value handling:** None required – the UCI HAR feature set has 0 missing values.
- **Encoding:** The activity label column was label-encoded for use only in external validation; it played no role in the clustering algorithms themselves.
- **Feature scaling:** All 561 features were standardized using `StandardScaler` (zero mean, unit variance). Standardization is essential here since K-Means and DB-SCAN both rely on Euclidean distance, and the 561 features (raw sensor statistics such as mean, std, energy, correlation) are on very different natural scales.
- **Train-test split:** Not applicable – as an unsupervised task, clustering was performed on the full standardized 10,299-sample dataset rather than a train/test split.
- **Dimensionality reduction (for visualization only):** A 2-component PCA projection was used to visualize all three algorithms' cluster assignments in 2D, and a t-SNE projection (computed on a 50-component PCA-reduced representation, for computational feasibility) was used as a second, non-linear visualization for K-Means.

6 Machine Learning Algorithms

K-Means Clustering:

- **Working principle:** K-Means partitions the data into k clusters by iteratively assigning each point to its nearest centroid and then updating each centroid as the mean of its assigned points, minimizing within-cluster variance.
- **Mathematical formulation:** Minimizes the within-cluster sum of squares $J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$, where μ_i is the centroid of cluster C_i .
- **Advantages:** Simple, fast, and scales well to large datasets; produces compact, easily interpretable spherical clusters.
- **Limitations:** Requires k to be chosen in advance, assumes roughly spherical, similarly-sized clusters, and is sensitive to centroid initialization and outliers.

DBSCAN:

- **Working principle:** DBSCAN groups together points that are closely packed (density-connected), using a neighbourhood radius `eps` and a minimum point count `min_samples`; points in low-density regions are labelled as noise (-1) rather than forced into a cluster.
- **Mathematical formulation:** A point p is a *core point* if its `eps`-neighbourhood contains at least `min_samples` points; clusters are formed by connecting core points that are density-reachable from one another, and border/noise points are assigned accordingly.
- **Advantages:** Does not require the number of clusters to be specified in advance, can discover arbitrarily-shaped clusters, and explicitly identifies noise/outlier points.
- **Limitations:** Sensitive to the choice of `eps` and `min_samples`, struggles with clusters of varying density, and can be computationally expensive in high-dimensional spaces (561 raw features here).

Hierarchical (Agglomerative) Clustering:

- **Working principle:** Agglomerative clustering starts with every point as its own cluster and repeatedly merges the closest pair of clusters (per a linkage criterion) until the desired number of clusters remains, producing a hierarchy that can be visualized as a dendrogram.
- **Mathematical formulation:** With Ward's linkage, at each step the pair of clusters is merged that minimizes the increase in total within-cluster variance, $\Delta = \sum_{x \in C_i \cup C_j} \|x - \mu_{ij}\|^2 - \sum_{x \in C_i} \|x - \mu_i\|^2 - \sum_{x \in C_j} \|x - \mu_j\|^2$.
- **Advantages:** Does not require k to be fixed upfront (can be read off the dendrogram at any cut height), and Ward's linkage tends to produce compact, balanced clusters similar in spirit to K-Means.
- **Limitations:** Computationally expensive for large datasets (the full linkage/dendrogram computation scales poorly with n), and once two points are merged into a cluster this decision cannot be undone (greedy, non-backtracking).

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	Windows
Hardware	

8 Hyperparameter Selection

8.1 K-Means – Elbow Method and Silhouette Score

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	3,272,857	0.3966
3	2,921,078	0.3137
4	2,781,602	0.1537
5	2,654,790	0.1340
6	2,577,180	0.1063
7	2,519,774	0.0767
8	2,465,379	0.0729

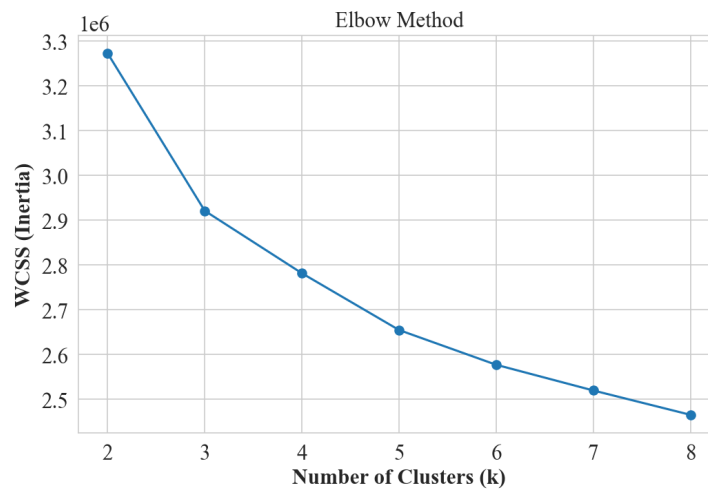


Figure 2: Elbow curve: WCSS vs. number of clusters k

Inference: WCSS decreases steadily and smoothly as k increases, without an unambiguous sharp elbow – the curve flattens gradually rather than bending abruptly at any single point, which is a common pattern for high-dimensional sensor data where structure exists at multiple granularities (static vs. dynamic activities, and within those, the individual activity types).

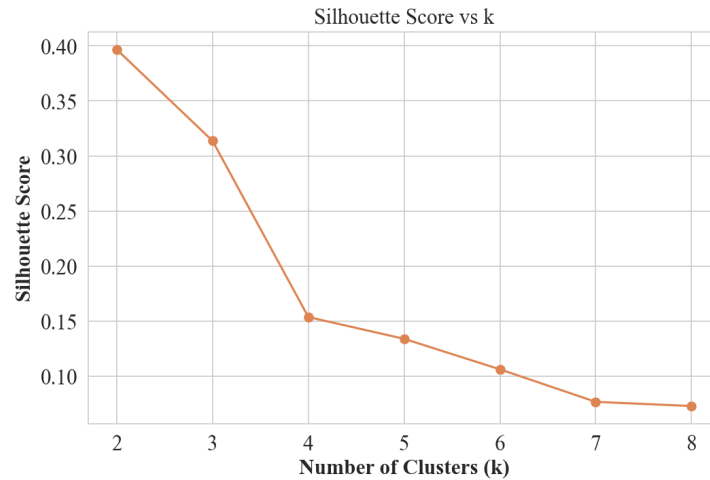


Figure 3: Silhouette score vs. number of clusters k

Inference: The silhouette score is highest at $k = 2$ (0.3966) and drops sharply thereafter, meaning that from a purely geometric standpoint the data separates most cleanly into two broad groups. This is consistent with the well-known coarse structure of the HAR feature space: static postures (SITTING, STANDING, LAYING) versus dynamic locomotion (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS).

Selected k : Although the silhouette score peaks at $k = 2$, the **true number of activities is 6**, and $k = 2$ would only recover the coarse static/dynamic split rather than the individual activities. To allow a meaningful comparison against the six ground-truth activity labels via external metrics (ARI, NMI), $k = 6$ was selected as the final K-Means configuration, matching the known number of underlying classes; the trade-off (lower silhouette score at $k = 6$ than at $k = 2$) is discussed in Section 11.

8.2 DBSCAN – Parameter Selection

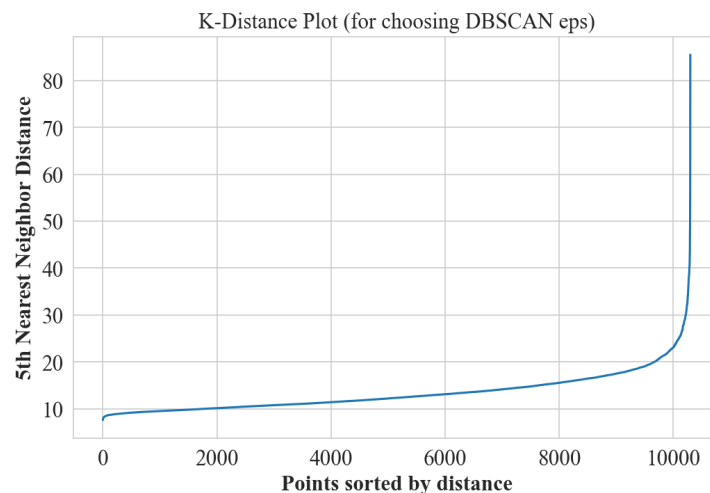


Figure 4: K-distance plot (5th nearest-neighbour distance, sorted) used to guide ϵ selection

Inference: The k-distance plot shows a gradual rise with a knee region in the middle-to-upper range of sorted distances rather than one sharp, unambiguous elbow, reflecting

the high dimensionality (561 standardized features) of the raw feature space used for DBSCAN. Based on this plot and iterative testing, **eps = 15** and **min_samples = 10** were selected as the final DBSCAN configuration, applied directly on the full standardized 561-feature data (not the 2D PCA projection, which is used only for visualization).

Parameter	Value
eps	15
min_samples	10
Clusters found	5 (plus noise)
Noise points	2,398 out of 10,299 (23.3%)

8.3 Hierarchical Clustering – Linkage and k

Parameter	Value
Linkage criterion	Ward
Number of clusters	6 (matched to K-Means and the true activity count for comparability)

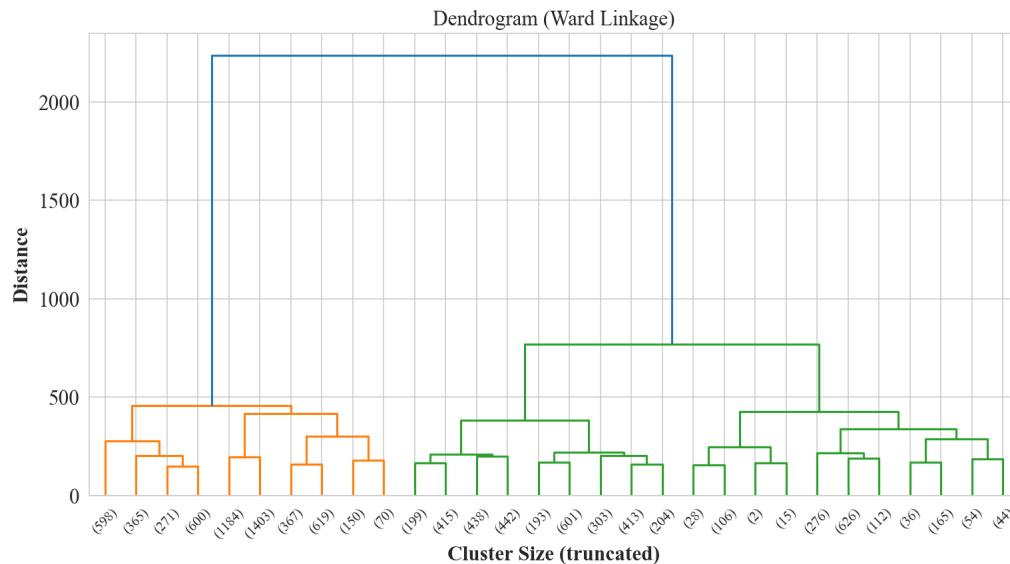


Figure 5: Ward-linkage dendrogram (truncated to the last 30 merges)

Inference: The dendrogram shows a clear early split into two major branches at a large distance, consistent with the static/dynamic separation seen in the silhouette analysis, with finer sub-clusters emerging only at much smaller merge distances within each branch. This tree structure visually confirms that the six individual activities are not equally distinct from one another – some activity pairs merge much later (i.e. are more similar) than others.

9 Results

9.1 Internal Validation Metrics

Algorithm	Silhouette	Davies–Bouldin	Calinski–Harabasz
K-Means ($k = 6$)		0.1063	2.3836
DBSCAN (eps=15, min_samples=10)	0.2280	1.1543	50.65
Hierarchical (ward, $k = 6$)		0.1110	2.4820
			2,349.67

9.2 External Validation Metrics (against true activity labels)

Algorithm	Adjusted Rand Index (ARI)	Normalized Mutual Information (NMI)
K-Means ($k = 6$)		0.4196
DBSCAN (eps=15, min_samples=10)		0.0800
Hierarchical (ward, $k = 6$)	0.4599	0.6015

Best overall model: Hierarchical clustering (Ward linkage, $k = 6$) achieves the best agreement with the true activity labels (highest ARI and NMI), narrowly ahead of K-Means, while DBSCAN – despite having the best internal Silhouette and Davies–Bouldin scores – has by far the weakest external agreement.

10 Result Visualization

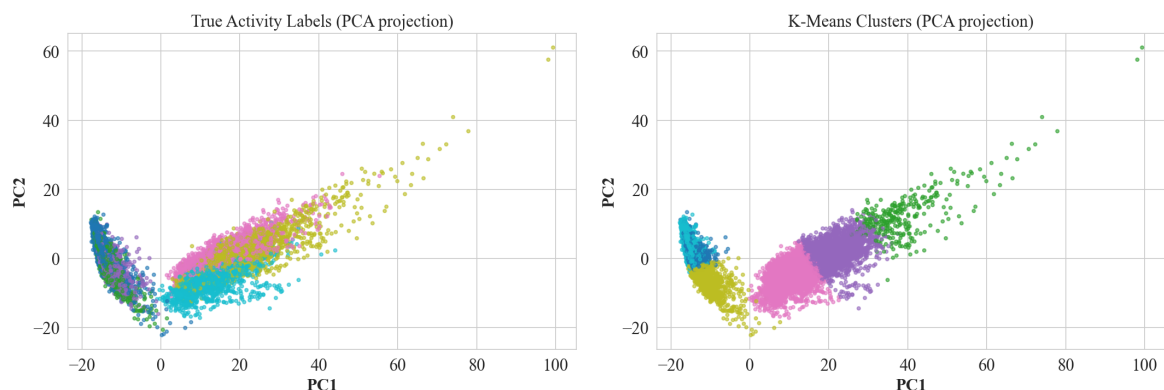


Figure 6: PCA (2D) projection coloured by true activity labels (left) and by K-Means ($k = 6$) cluster assignment (right)

Inference: The true-label plot shows the six activities forming two broad arcs (static activities clustered tightly on the left, dynamic activities spreading in a fan shape on the right), and the K-Means clusters visibly follow this same overall shape, correctly separating most of the static-vs-dynamic structure, though the boundaries between individual dynamic activities (the three walking variants) are less crisp than the boundary between static and dynamic groups.

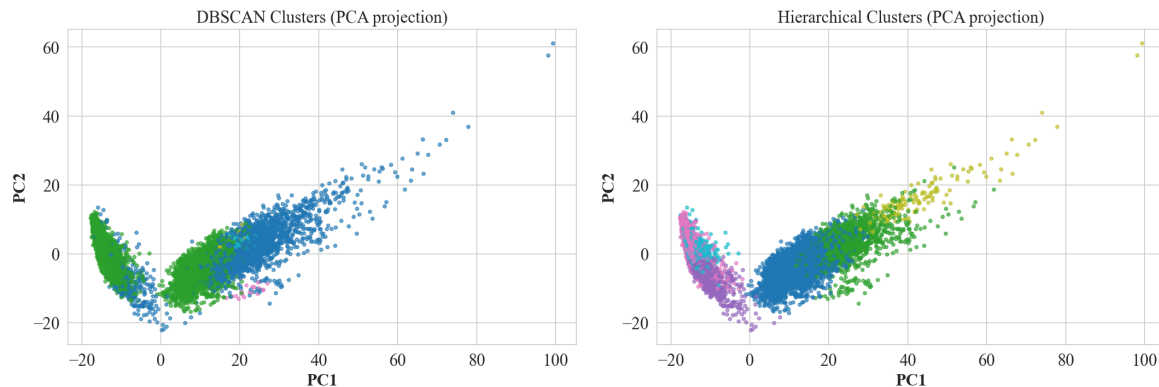


Figure 7: PCA (2D) projection coloured by DBSCAN cluster assignment (left) and Hierarchical (Ward) cluster assignment (right)

Inference: DBSCAN produces a small number of dense, tightly-bounded clusters concentrated mostly in the static-activity region, while classifying a large fraction of the more spread-out dynamic-activity points as noise (visible as the diffuse light-coloured scatter) – this explains its high internal Silhouette score (which only credits dense, well-separated clusters) alongside its very low external ARI/NMI (since it fails to assign most points to a meaningful cluster at all). Hierarchical clustering’s cluster boundaries closely resemble K-Means’, consistent with the two producing similar ARI/NMI scores.

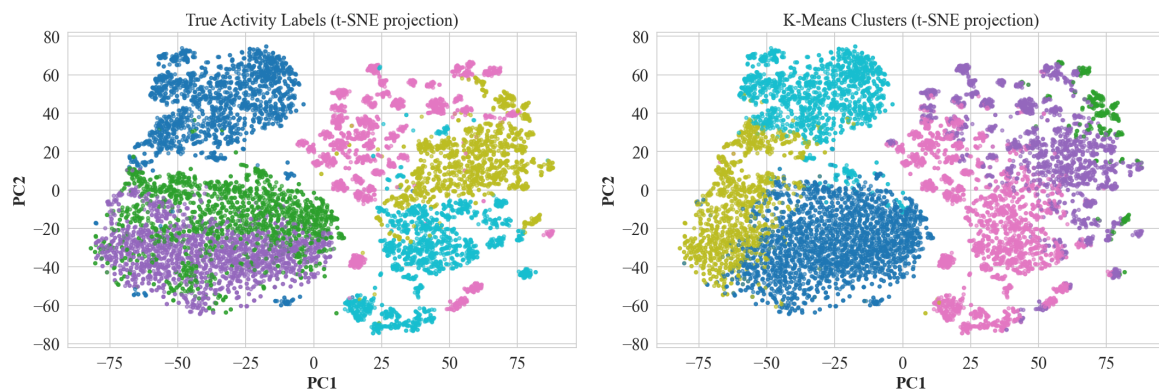


Figure 8: t-SNE projection coloured by true activity labels (left) and K-Means cluster assignment (right)

Inference: The non-linear t-SNE projection separates the six true activities into more distinct, compact groups than the linear PCA projection did, revealing structure among the dynamic activities that PCA compressed together. The K-Means cluster assignment captures some, but not all, of this finer six-way structure – consistent with K-Means’ moderate (not high) ARI/NMI scores, since k -means clusters are constrained to be roughly spherical in the original 561-dimensional space, which does not perfectly align with the non-linear manifold that t-SNE reveals.

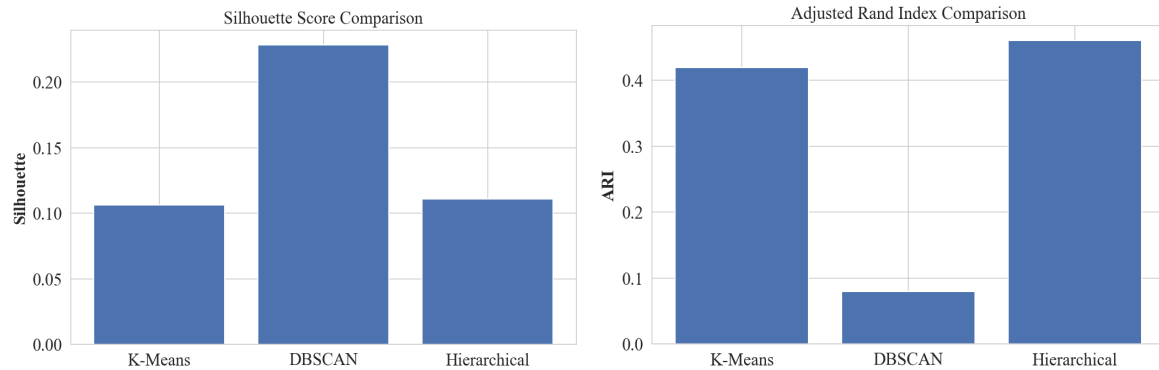


Figure 9: Silhouette score comparison (left) and Adjusted Rand Index comparison (right) across all three algorithms

Inference: The two panels tell opposite stories: DBSCAN wins decisively on the internal Silhouette metric, while Hierarchical (and K-Means close behind) wins decisively on the external ARI metric. This divergence is the central finding of this experiment – an algorithm that looks best by a purely geometric/internal measure is not necessarily the one that best recovers the true semantic (activity) structure, especially when that algorithm (DBSCAN) discards a large fraction of points as noise rather than assigning them to any cluster.

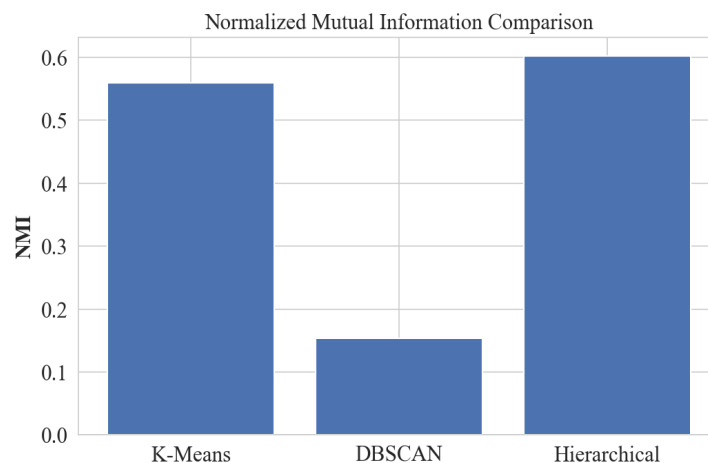


Figure 10: Normalized Mutual Information comparison across all three algorithms

Inference: The NMI comparison mirrors the ARI comparison, with Hierarchical clustering again narrowly ahead of K-Means and both far ahead of DBSCAN, reinforcing that Ward-linkage hierarchical clustering and K-Means (both of which produce compact, roughly-equal-sized partitions) align better with the six human-defined activity categories than DBSCAN’s density-based, noise-excluding approach does on this dataset.

11 Observations and Comparative Analysis

- **Which algorithm produced the most meaningful clusters?** By external validation (ARI/NMI) against the true activities, **Hierarchical (Ward) cluster-**

ing produced the most meaningful clusters, marginally ahead of K-Means; both substantially outperformed DBSCAN on this criterion.

- **How sensitive was K-Means to the choice of k ?** Very sensitive: the silhouette score fell from 0.397 at $k = 2$ to just 0.106 at $k = 6$, a more than $3\times$ drop, showing that the data's cleanest geometric separation (static vs. dynamic) is coarser than the six semantic activity classes actually present.
- **Did DBSCAN detect noise or small clusters effectively?** Yes – DBSCAN flagged 23.3% of points (2,398 of 10,299) as noise and found 5 clusters, achieving the best internal Silhouette (0.228) and Davies–Bouldin (1.154) scores of the three algorithms by concentrating on dense cores and excluding ambiguous, lower-density points. However, this same behaviour is why its external agreement with the true activities (ARI = 0.080) was much weaker – it is effective at finding *dense regions*, not necessarily at recovering all *semantic categories*.
- **How does linkage choice affect hierarchical clustering?** Ward's linkage was used throughout this experiment specifically because it minimizes within-cluster variance similarly to K-Means, producing compact, comparably-sized clusters; other linkage criteria (single, complete, average) were not run in this notebook but are generally known to be more sensitive to noise and chaining effects (single linkage) or to produce more uneven cluster sizes (complete/average), which would likely reduce agreement with the fairly balanced six-activity ground truth.
- **Which internal metric best matched visual intuition?** The Calinski–Harabasz index (highest for K-Means at 2,556.5, close behind for Hierarchical at 2,349.7, far lower for DBSCAN at 50.6) matched the PCA/t-SNE visual intuition most closely – among K-Means and Hierarchical, both of which visually produce well-separated, globular clusters in the PCA/t-SNE plots – better than the Silhouette score, which favoured DBSCAN's smaller, noise-excluding clusters that looked visually less representative of the overall data structure.

12 Conclusion

This experiment applied K-Means, DBSCAN, and Hierarchical (Ward) clustering to the 561-feature UCI HAR dataset and compared them via elbow/silhouette analysis, a k -distance-guided DBSCAN parameter search, and both internal and external validation metrics. Hierarchical (Ward) clustering achieved the best agreement with the true six-activity structure (ARI = 0.460, NMI = 0.601), narrowly ahead of K-Means ($k = 6$; ARI = 0.420, NMI = 0.559), while DBSCAN, despite winning on purely internal geometric metrics (Silhouette = 0.228, Davies–Bouldin = 1.154), substantially underperformed both on external agreement (ARI = 0.080) due to labelling a large fraction of points as noise. This experiment illustrates a key lesson in unsupervised learning: the algorithm judged "best" depends heavily on which validation criterion (internal vs. external) is prioritized, and a clustering method's ability to find dense, well-separated regions does not guarantee that those regions correspond to the categories a human observer (or ground-truth label set) would define.

13 Key Learning Outcomes

- Applying the elbow method and silhouette score to select k for K-Means, and recognizing when the geometrically "best" k (here, 2) differs from the semantically meaningful number of classes (here, 6).
- Using a k-distance plot to guide DBSCAN's `eps` parameter, and interpreting DBSCAN's noise-point mechanism.
- Comparing linkage-based hierarchical clustering against K-Means via a dendrogram and cluster assignments.
- Distinguishing internal (Silhouette, Davies–Bouldin, Calinski–Harabasz) from external (ARI, NMI) cluster-validation metrics, and recognizing that they can rank algorithms very differently.

14 References

1. D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "A Public Domain Dataset for Human Activity Recognition Using Smartphones," *ESANN*, 2013.
2. J. MacQueen, "Some Methods for Classification and Analysis of Multivariate Observations," in *Proc. 5th Berkeley Symp. on Mathematical Statistics and Probability*, 1967, pp. 281–297.
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4. F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.